

Abstraction Project

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Proving Output Constraints with Symbolic Simulation

We only need to focus on proving positive output constraints, i.e. that under a certain condition, a signal will be high.

In fact, with the formulation of relational properties, we don't even need to deal with specific conditions, but our proof will deal with the general case.

Symbolic Simulation Invariant

Consider some residual $Q[X, C]$ (the `highexpr` in a symbolic simulation evaluation sequence) in a signal $s(C, T)$ at a fixed time step. We always have the fact that

$$Q[X, C] \wedge R[X, C, T] \rightarrow s(C, T)$$

(For all X, C, T)

This can be proved via induction on the fan-in of S .

The basecase is where s is an input or state variable corresponding to a certain time step, with `highexpr` P . In this case

$$Q = P^R = P_R \wedge \forall T (R[X, C, T] \rightarrow P[C, T])$$

This directly gives us that for all X, C, T where $Q[X, C] \wedge R[X, C, T]$, we have $P[C, T]$. But $s(C, T) = P[C, T]$ so we are done.¹

An inductive case example:

Suppose $s = s_1 \wedge s_2$. We have Q_1 and Q_2 as the residuals of s_1 and s_2 respectively.

- $Q_1[X, C] \wedge R[X, C, T] \rightarrow s_1(C, T)$
- $Q_2[X, C] \wedge R[X, C, T] \rightarrow s_2(C, T)$
- So $Q_1[X, C] \wedge Q_2[X, C] \wedge R[X, C, T] \rightarrow s_1 \wedge s_2$
- But is equivalent to $Q[X, C] \wedge R[X, C, T] \rightarrow s$ since $Q = Q_1 \wedge Q_2$ via the symbolic simulator

Proof of Correctness for Output Constraint Checking

Output constraints that we wish to prove are of the form:

$$\forall C \forall T (P[T, C] \rightarrow s(C, T))$$

If we have that $P_R[X, C] \rightarrow Q[X, C]$ and the coverage condition $\forall T \forall C \exists X R[X, C, T]$ holds, then this will be true.

First remember that $P_R = \exists T (R[X, C, T] \wedge P[C, T])$.

Fix some C, T such that $P[C, T]$ is true.

- Since $P[C, T]$ is true, then $\forall X (R[X, C, T] \rightarrow P_R[X, C])$ by the definition of P_R .
- Consider some X^* such that $R[X^*, C, T]$ is true, this must exist by the coverage condition
- Since $P[T, C]$ and $R[X^*, C, T]$ are true, then we must have $P_R[X^*, C]$
- But now since $P_R[X, C] \rightarrow Q[X, C]$, we have $Q[X^*, C]$ as well

¹Note that we don't use the weak preimage part. Indeed, the analysis will still work, but the weak preimage serves to remove useless/inconsistent indexing cases. This will help in avoiding false counter examples.

- Using the symbolic simulation invariant, since $Q[X^*, C]$ and $R[X^*, C, T]$ are true, then $s(C, T)$ is true
- Since this did not rely on the values of C, T , then we have that $\forall C \forall T (P[T, C] \rightarrow s(C, T))$ is true

Lowexpr Values

We can set up an analogous invariant for the lowexpr values in the symbolic simulation. We will then be able to prove that if L is the lowexpr of $s(C, T)$, then $(L[X, C] \wedge R[X, C, T] \rightarrow \neg s(C, T))$.

We can then prove output constraints of the form $\forall C \forall T (P[T, C] \rightarrow \neg s(C, T))$ in a similar manner to the highexpr case.

For our purposes where we use just need to prove that a certain wire is always high, we don't need this component. That being said, analysis of $L[X, C]$ is important for finding unproven cases and counter examples.